

Human-AI Problem Solving

Adapted from George Pólya, How to Solve It, 2nd ed. (Princeton University Press, 1957).

Don't use AI just to get the final answer. Instead, use Pólya's four-step structure to make AI a collaborator in your problem solving. The questions below can guide both your own reasoning and the prompts you give AI—helping you understand the problem, develop and test a plan, carry it out carefully, and reflect on the result.

1 • Understand the Problem

- What is the unknown? What data are you given? What conditions must the answer satisfy?
- Can the conditions be satisfied? Are they sufficient, insufficient, redundant, or contradictory?
- Can you draw a figure or introduce some useful notation to make the relationships clearer?
- Can you separate the conditions into parts and write each one down?

2 • Devise a Plan

- How are the data and the unknown connected?
- What smaller or auxiliary problem could help you link the data and the unknown?
- Have you seen this problem before, or a similar problem in a different context?
- Can you think of a familiar problem with the same or similar unknown?
- Can you think of a problem you have seen before and use its result or method, perhaps by adding an auxiliary element?
- Can you state your problem differently; or use the definitions?
- Can you think of a more accessible, more general, more special, or an analogous problem? Can you solve part of the problem, or adjust the data or the unknown so they are easier to connect?
- Does your plan use all the data, the full set of conditions, and every essential idea in the problem?

3 • Carry Out the Plan

- Check each step: can you see clearly why it is correct?
- Can you justify or generate a proof that the step is correct?

4 • Look Back

- Can you check the result? Can you check the reasoning?
- Can you derive the solution in a different way?
- Can you now see the solution at a glance?
- Can you use this result or method to solve another problem?